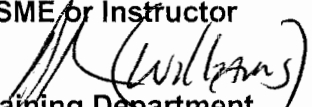
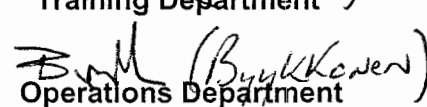


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM 1 & 2		
SYSTEM:	Loss of Control Air		
TASK:	TCAF a Loss of Control Air		
TASK NUMBER:	N1140070401		
JPM NUMBER:	11-01 ILOT Sim g		
ALTERNATE PATH:	☒	K/A NUMBER:	APE 065 AA2.06
APPLICABILITY:		IMPORTANCE FACTOR:	3.6 4.2
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	S2.OP-AB.CA-0001, Rev. 17 Loss of Control Air S2.OP-AR.ZZ-0011, Rev. 58, pages 118-128		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	8 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-4-12
Validated By:	Brennan SME or Instructor	Date:	8-24-12
Approved By:	 Training Department	Date:	9/12/12
Approved By:	 Operations Department	Date:	9/5/12
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Loss of Control air

TASK: TCAF Loss of Control Air

TASK NUMBER: N1140070401

SIMULATOR SETUP:

IC-252

RT-1

MALF: CA0221 #2 STATION AIR CMPRSSR TRIP

REM : CA13D LOCKOUT #1 STATION AIR CMPRSSR, Delay 1:30

I/O : AQ01 OVLO COMP 1 TROUBLE ALARM, Fin Val: ON, Delay 1:30

Other setup:

REM: CA15D LOCKOUT #3 STATION AIR CMPRSSR

CA11D #1 EMERGENCY AIR COMPRSR IN AUTO Fin Val: MANUAL

I/O : AQ01 OVLO COMP 3 TROUBLE ALARM, Fin Val: ON

INITIAL CONDITIONS:

- Unit 2 is operating at 100% power.
- A WG release is in progress from 21 GDT.
- #3 SAC is C/T for scheduled maintenance.
- Unit 1 is operating at 100% power.

INITIATING CUE:

You are the Unit 2 Reactor Operator. Respond to all indications and alarms.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Isolate Letdown
2. Terminate Gaseous Release
3. Trip the Rx prior to auto trip on lo lo SG NR level.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note: The #1 and #3 Compressor Trouble alarms on 2CC1 do NOT have audible or flash capabilities (S2.OP-AR.ZZ-0011 page 120).		
		Simulator Operator: Insert <u>RT-1</u> after operator assumes watch.			
			Determines #2 Station Air Compressor (SAC) has tripped, and the Unit 2 Emergency Control Air Compressor (ECAC) has automatically started. Note: All 3 SAC supply breakers open is an auto start signal for both ECAC's. The Unit 2 ECAC will auto start, (Unit 1 will not). By design, the ECAC will not load until its control air header pressure lowers to 85 psig. 2A control air header is supplied from #2 ECAC, 2B control air header is supplied from #1 ECAC.		
			Note: The next-to-load SAC (follow) will not immediately start. It requires Station Air header pressure to drop to 5 psig below the follow setpoint (105 psig) for ~ 5 seconds.		
			Refers to the S2.OP-AR.ZZ-0011, Control Console 2CC1, ARP for COMPR 2 TROUBLE.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	ARP 3.1	<u>IF</u> SAC 2 trips, <u>THEN</u> : A. COORDINATE (as necessary) with Unit 1 B. ENSURE operating <u>OR</u> START at least one SAC IAW SC.OP-SO.SA-0001(Z), Station Air System Operation.	Coordinates with Unit 1 to start at least one SAC IAW SC.OP-SO.SA-0001(Z), Station Air System Operation. Cue when contacted as Unit 1: Unit 1 will ensure #1 SAC is placed in service.		
	ARP 3.2	<u>IF AT ANY TIME</u> Station Air pressure cannot be maintained ≥ 100 psig, <u>THEN</u> GO TO S2.OP-AB.CA-0001(Q), Loss of Control Air.	GOES TO S2.OP-AB.CA-0001(Q), Loss of Control Air, when Station Air pressure lowers to 100 psig if AB not entered previously.		
			Simulator Operator: Ensure #1 SAC locks out 1:30 after insertion of RT-1, then announce twice on plant page: "#1 Station Air Compressor Trip."		
			Recognizes that no SAC's are running, and enters S2.OP-AB.CA-0001 if not entered previously.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note: Approximate event times are listed below <u>from insertion of RT-1</u>. 1 min – 1 SAC auto starts. 1 min 30 sec – 1 SAC trips. 3 min 50 sec – 2 ECAC begins loading. 4 min 10 sec - 2B Control Air header pressure reaches 80 PSIG, and SA low pressure alarm. 4 min 25 sec – CA low pressure alarm. 6.5 min – 21-24BF19 Feed Reg Valves begin closing. 7 min 10 sec – G-15 OHA ADFCS TRBL.		
	AB.CA 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary. Cue IF Required: If CAS Step 8.0 action is being taken to refer to Attachment 2 for valve fail positions and for further action to be taken, then: Cue: The CRS will refer to Attachment 2		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air
TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.2	Do either of the following conditions exist? - All Station Air Compressors stopped <u>OR</u> - Either Station Air Header <100 psig	Determines all SAC's are stopped or determines Station Air Header pressure is <100 psig.		
	3.3	IF a loss of 1F Group Bus has occurred, AND 1H Group Bus is available, THEN SEND an Operator to restore power to #1 SAC Auxiliary Oil pump by transferring 11 Turbine West 230V Control Center IAW Attachment 14, Section 1.0.	Determines a loss of 1F Group Bus has not occurred. Cue if required: Unit 1 reports 1F Group Bus is energized.		
	3.4	IF a loss of 2F Group Bus has occurred, AND 2H Group Bus is available, THEN SEND an Operator to restore power to #2 SAC Auxiliary Oil pump by transferring 21 Turbine West 230V Control Center IAW Attachment 14, Section 2.0.	Determines a loss of 2F Group Bus has not occurred.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Loss of Control Air**

TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.5	IF a loss of 1H Group Bus has occurred, AND 1F Group Bus is available, THEN SEND an Operator to restore power to #3 SAC Auxiliary Oil pump by transferring 11 Turbine East 230V Control Center IAW Attachment 14, Section 3.0.	Determines a loss of 1H Group Bus has not occurred. Cue if required: Unit 1 reports 1H Group Bus is energized.		
	3.6	START or have Unit One Control Room START the Next to Load Station Air Compressor.	Determines no SAC's are available. Cue if required: If Unit 1 is contacted, report that #3 Station Air Compressor is tagged out and #1 Station Air Compressor has tripped, and that an operator has been dispatched to the Station Air Compressors.		
	3.7	Is either Station Air Header <100 psig?	Determines both Station Air Headers are <100 psig.		
	3.8	Attempt to START , or have Unit One Control Room START , the remaining Station Air Compressor.	Determines no SAC's can be started.		
	3.9	NOTIFY the SM/CRS to evaluate use of temporary air compressors.	Cue: CRS will evaluate use of temporary air compressors.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.10 3.11	Is 2A Control Air Header ≤ 88 psig?	Determines 2A control air header pressure is: ≤ 88 psig and determines #2 ECAC is in service OR >88 psig.		
	3.12 3.13	Is 2B Control Air Header ≤ 88 psig?	Determines 2B control air header pressure is: ≤ 88 psig and contacts Unit 1 to place #1 ECAC in service OR >88 psig. Cue: When Unit 1 is contacted to start #1 ECAC, state: " # 1 ECAC has tripped. "		
	3.14	SEND operators as necessary to: • LOCATE AND ISOLATE any air system leaks. • INVESTIGATE cause of air compressor trip/trouble.	Dispatches operators to: • LOCATE AND ISOLATE any air system leaks. • INVESTIGATE cause of air compressor trip/trouble.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.15	IF Station Air is lost, <u>AND</u> either of the following are in service: - Spent Fuel Pool Transfer Pool Weir Gate Seals - SG Nozzle Dams <u>THEN</u>: A. EVACUATE personnel from the areas subject to leakage/flooding. B. INITIATE monitoring of Fuel Pool and/or Reactor Vessel levels for the need to makeup due to leakage.	Determines Station Air is lost, and <u>neither</u> of the following are in service: - Spent Fuel Pool Transfer Pool Weir Gate Seals - SG Nozzle Dams		
	3.16	IF cooling water to the Station or Emergency Air Compressors is lost, <u>THEN</u> INITIATE the applicable portion of Attachment 6, Abnormal Cooling Water Lineups.	Determines cooling water to the Station or Emergency Air Compressors is not lost.		
	3.17	Are both 2A and 2B Control Air Headers <80 psig?	Determines both 2A and 2B Control Air Headers are not <80 psig.		
	3.18	Is 2B Control Air Header <80 psig?	Determines 2B Control Air Header is <80 psig.		
	3.55	CONTINUE			

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	3.56	ISOLATE Letdown by closing the following valves: A. 2CV3, 45 GPM ORIFICE B. 2CV4, 75 GPM ORIFICE C. 2CV5, 75 GPM ORIFICE D. 2CV2 E. 2CV277 F. 2CV7	Isolates Letdown by closing the following valves: A. 2CV3, 45 GPM ORIFICE B. 2CV4, 75 GPM ORIFICE C. 2CV5, 75 GPM ORIFICE D. 2CV2 E. 2CV277 F. 2CV7		
*	3.57	VERIFY any Liquid or Gaseous release is STOPPED by ensuring the following valves are closed: • 2WL51, TO CIRC WTR DISCHARGE • 2WG41, GAS DECAY TK TO PLANT VENT	Determines 2WG41, GAS DECAY TK TO PLANT VENT is open, and shuts 2WG41, GAS DECAY TK TO PLANT VENT. Verifies 2WL51 is shut.		
*	CAS	<u>IF AT ANY TIME</u> Station Air is lost, <u>THEN:</u> • INITIATE monitoring of 21-24BF19 operation. • IF any BF19 goes CLOSED, AND the applicable SG water level cannot be maintained, THEN TRIP the Reactor, CONTINUE with this procedure AND GO TO 2-EOP-TRIP-1, Reactor Trip Or Safety Injection.	Identifies all BF19 valves closing. Trips the Reactor. CUE IF REQUIRED: IF reactor is tripped prior to completion of steps 3.56 AND 3.57, then allow immediate actions for the Rx trip to be performed. THEN CUE: The CRS and another RO will continue performing TRIP-1. You are to continue performing AB.CA.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Terminate JPM once Rx has been tripped, Letdown isolated, and the WG release terminated.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- Y 1. Task description and number, JPM description and number are identified.
- Y 2. Knowledge and Abilities (K/A) references are included.
- Y 3. Performance location specified. (in-plant, control room, or simulator)
- Y 4. Initial setup conditions are identified.
- Y 5. Initiating and terminating Cues are properly identified.
- Y 6. Task standards identified and verified by SME review.
- Y 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- Y 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 17 Date 8-24-12
- Y 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- N/A 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: R R Brennan

Date: 8-24-12

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

- Unit 2 is operating at 100% power.
- A WG release is in progress from 21 GDT.
- #3 SAC is C/T for scheduled maintenance.
- Unit 1 is operating at 100% power.

INITIATING CUE:

You are the Unit 2 Reactor Operator. Respond to all indications and alarms.